

Comparative Analysis of AI Adoption: Evidence from Indonesia

1st Dionisius Andre Kusuma Dewa

Department of Economics

University of Michigan

Ann Arbor, United States of America

dioandre@umich.edu

Abstract—Artificial Intelligence (AI) adoption has been growing in recent years. Impact on productivity gain that comes from automation and task complementarities are being observed in the US. However, the rate of adoption of AI in the economy across countries has not been clear. This paper will focus on Indonesia and its adoption rate of AI with usage of Claude AI as a proxy. This study will compare the usage of AI across tasks, occupations and country level, comparing it to the overall global AI usage by borrowing theory of comparative analysis and measure the Revealed Comparative Advantage (RCA) of AI Adoption in Indonesia. The paper will also examine the trajectory of AI adoption using the Spearman to see whether the adoption in a country on a occupation level has no dependence with the AI adoption and observed exposure in the US. This will help us framed on how AI is being used on different task and occupations and also across countries in the globally.

Index Terms—AI adoption, Indonesia, labor market, revealed comparative advantage, Anthropic Economic Index

I. INTRODUCTION

The growth in capabilities shown by Artificial Intelligence (AI) in recent years has spurred an ongoing debate over whether human labor will be replaced by AI. Reference [3] have argued that artificial intelligence will likely persist, but simply displacing labor would not yield good results for the labor market, and there is no inevitable future of AI development in which we cannot advocate. The AI's impact on labor displacement would likely differ across industries.

Reference [1] made claims that the AI impact on productivity can come from automation and task complementarities of labor as these are the two out of four possible channels in the study. The adoption of AI in a country's production function would happens through this increase of productivity. However, the rate of adoption of AI in the economy has not been clear.

Anthropic recently released a report titled "Anthropic Economic Index Report: Uneven Geographic and Enterprise AI Adoption" [4]. The report introduce measures to gauge AI usage, adding new information to the research on AI and how it shapes current and future labor market. Understanding the risk incurred with the increasing speed of AI adoption across industry and jobs are essential for the business and countries. Labor across the world will need to adapt and get

the optimal benefits while minimizing the potential risks of negative outcome.

We argue that we can compare the rate of adoption between countries through a comparative analysis. This study focus on comparative analysis of AI adoption that can be done through estimating the usage of AI in a task-level and occupation level and country level, comparing it to the overall global AI usage. We would apply the Revealed Comparative Advantage as the comparative analysis method [5] and use it to measure the relative usage of AI in a certain occupation in a country to the global median usage.

This paper will focus on Indonesia and the its adoption rate of artificial intelligence with usage of Claude AI as a proxy. AI adoption in Indonesia will be examined through its usage across different occupations. The data provided will help us understand the current state of AI adoption in Indonesia.

Massenkoff, M., & McCrory, P suggested a new measure of labor displacement risk in US, called observed exposure by measuring the current AI usage in a certain tasks and occupation, observed its level of automation and compare it to the theoretical capabilities of AI usage developed by [6]. Such observed measure could give information on a trajectory of AI adoption in the US and how it currently moves across tasks and occupations.

While observed exposure on a country level for other countries have not been studied, this study will also examine the trajectory of AI adoption in Indonesia by comparing the RCA AI Adoption with the constructed AI observed exposure in the United States of America (US) studied by [8]. This would reveal the independence of AI adoption in Indonesia to the US adoption and occupation exposure rate.

II. LITERATURE REVIEW

Reference [2] define AI-based productivity gains can come from automation, task complementarity, deepening of automation, and new tasks created to accommodate AI use in workplace. Automation is the model of AI taking over and reduce the time needed to perform task. Task complementarity is the use of AI through the increase of productivity of labor in performing tasks that are not fully automated. Details and about the deepening of automation and new tasks created to accommodate AI use in workplace is out of the scope of this paper.

Reference [1] estimates that the impact of AI in the production function happens through the cost savings effects either by the automation or task complementarities. These increase in the productivity would only affect the portion of the GDP with tasks that has exposure to the AI adoption in its production.

Impact of Artificial Intelligence in the form of Large Language Model and Generative Pre-trained Transformer (GPT) on the labor market was examined in the paper of [6]. The paper use data on activities and tasks performed by occupation in the use in the O*NET database which contain information on 1,016 occupations and its respective Detailed Work Activities (DWA) and task. Task is defined as an occupation-specific unit of work that may be associated with zero, one, or multiple DWAs. The study by [6] used GPT-4 to classify all 19,265 tasks and 2,087 DWA to produce a task-level exposure estimate. It used both human and GPT-4 generated annotations to create estimates of exposure, measured as β , which measure whether it is theoretically possible for an LLM to make a task at least twice as fast.

Reference [4] released the Anthropic AI Usage Index (AUI) to map how AI has been used across countries. AUI captures the usage of Claude relative to its working age population. This reveals how a country's AI usage patterns compared to other country. The AUI however, does not track on task level and occupation level.

Reference [5] find that Revealed Comparative Advantage (RCA) can helps explain the difference of production function across countries through the export as a result of the production. RCA works by evaluating the relative level of an export to the total export of a country and also to the total global export.

$$RCA_{ij} = \frac{E_{ij}/E_{tot,j}}{E_{i,world}/E_{tot,world}} \quad (1)$$

- RCA_{ij} = country j 's revealed comparative advantage for commodity group i ;
- E_{ij} = exports of good i by country exporter j ;
- $E_{tot,j}$ = total exports of country j — merchandise alone, excluding services;
- $E_{i,world}$ = world exports of good i ;
- $E_{tot,world}$ = total world exports.

We argue that RCA can be used to do comparative analysis of AI usage in the tasks level and occupation level. The adoption of AI to the production function either through the automation or task complementarities and the difference of such adoption between countries can be estimated from the relative portion of the task and country level use to the total global use of AI. This will help understand how AI is currently being used in the workforce in each country and also across countries in the global level. The estimation will give information on relative adoption rate of AI to the production function of each country.

[8] suggested a new measure to calculate AI displacement risk called the observed exposure. The study measured the gap between theoretical capability of AI usage as studied in

[6] and request that are being carried in AI platform, in this case, Claude. These tasks will then be adjusted to how it is being utilized whereas automated use will receive full weight and task complementary receive half weight. Then, these tasks measures are averaged to the occupation level weight using portion of time needed to complete each task.

This paper will also examine how this observed exposure will relate to the relative AI adoption of a country through RCA. We argue that understanding the relationship between the RCA and observed exposure will help us understand whether AI adoption in a country on a occupation level have positive, negative, or no relation at all with the observed exposure that have been measured. The information is important to understand the relative trajectory of AI adoption in a particular country compared to observe exposure.

Measuring such relationship with data of observed exposure from [8] would required us to see the distribution of occupation level data. Since there are nonparametric data in observed exposure with zero value on certain occupation level data, we look to a non parametric method of test for independence between two variables.

Reference [7] find that the Spearman rank correlation coefficient is know as the product moment sample correlation of R and S , where both are the rank of n X observation from least to greatest and the rank of n Y, respectively.

$$r_s = \frac{12 \sum_{i=1}^n [R_i - \frac{n+1}{2}] [S_i - \frac{n+1}{2}]}{n(n^2 - 1)} \quad (2)$$

$$r_s = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)} \quad (3)$$

where $D_i = S_i - R_i$, $i = 1, \dots, n$. We will use a two-sided Spearman test for independence as follows:

$$\text{Reject } H_0 \text{ if } |r_s| \geq r_{s,\alpha/2}; \text{ otherwise, do not reject.} \quad (4)$$

III. METHODOLOGIES

The dataset use in this paper are as follow 1) The AI Usage Index (AUI) 2) The O*NET task classification and occupation 3) the observed exposure from [8].

We will first do data cleaning process and data distribution analysis. Geographic data from the AUI that has non identifiable country origin was filtered out. This will allow us to do a clear analysis on geographical data. We performed the distribution analysis to identify data normality.

RCA measurement is performed on the request clusters using the AUI index. This will help us identified which requests clusters are being carried more in Indonesia compared to global median.

We also use the SOC Major Group list employed by the US Bureau of Statistic (2018) to map task to SOC major groups. AUI on task level data are then being matched to these SOC level major group. RCA measurement is performed on the AUI at task level with the SOC main group and country level classification. Indonesia's RCA AI Adoption profile will then be built from the measured result. The results are compared to the calculated RCA AI Adoption global median.

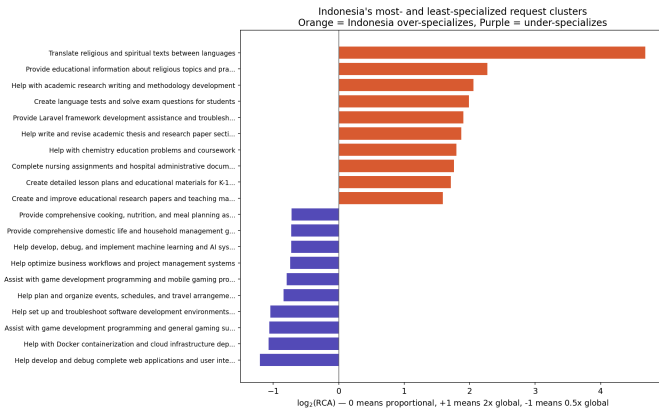


Fig. 1. Indonesia RCA on Request Clusters vs Global Median

Spearman test for independence is carried for the Indonesia RCA AI Adoption at the occupation level and the Observed Exposure data. Spearman test will be examined to see whether the AI adoption in Indonesia is independent of US Observed Exposure. Occupations will then be mapped into a matrix with 4 quadrants as follow: 1) Quadrants I with low observed exposure and $RCA > 1$; 2) Quadrant II with high observed exposure and $RCA > 1$; 3) Quadrants III with low observed exposure and $RCA < 1$; and also 4) Quadrant IV with high observed exposure and $RCA < 1$.

IV. RESULT

RCA on the requests have revealed how Indonesian uses AI on requests compared to global median. This request depict all tasks and activities beyond just work-related. This helped us framed on how AI is being used broadly across population in Indonesia.

Figure 1 shows the Indonesia RCA AI Adoption using the requests clusters. We can see that at the top of the requests clusters RCA is the "Translate religious and spiritual texts between languages" and "Provide educational information about religious topics" with the third highest RCA on request cluster is "Help with academic research writing and methodology development". These requests cluster RCA revealed cultural and occupational needs that are being adapted quicker in Indonesia. On the other hand, the under-specialized request cluster is "Help develop and debug complete web application", which tell us that some request around web developing are under utilized compared to the use in other countries.

Figure 2 shows the Indonesian RCA AI Adoption by occupation group. It can be seen that the first occupation group with highest RCA is the Educational Instruction with 2.47 and Life, Physical & Social Science next with 1.46. This revealed that the educational instruction still utilizes AI with higher used in comparison with global median. On the other hand, the least non-zero value of RCA is in the Sales occupation group with 0.32, Office & Admin Support with 0.44, Computer & Mathematical with 0.62 point. This revealed there are still under utilized AI adoption in Indonesia relates to the corporate category or white collar job.

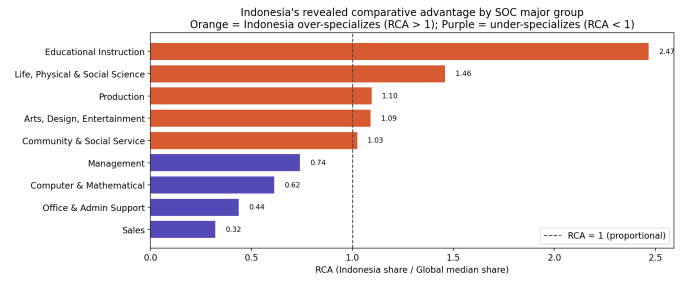


Fig. 2. Indonesia's RCA AI Adoption by Occupation Group

=== SPEARMAN CORRELATION MATRIX ===
n = 142, * p<0.05, ** p<0.01, *** p<0.001

	US exposure	Indonesia RCA	Indonesia share	Global median
US exposure	1.000	-0.001	+0.102	+0.313***
Indonesia RCA	-0.001	1.000	+0.923***	+0.489***
Indonesia share	+0.102	+0.923***	1.000	+0.688***
Global median	+0.313***	+0.489***	+0.688***	1.000

Spearman rho = -0.001, p = 0.9879

Fig. 3. Spearman Test for Independence

We then use the Spearman test for independence to see whether Indonesia's RCA AI Adoption is independent to the measured observed exposure in the US. The result are as follow: The Spearman test result shows that there is statistically significant association of US exposure to the global median AI usage. It also shows that there is no statistically significant evidence of Indonesia's RCA have association to the observed exposure of US. This means that Indonesia AI adoption have different adoption pattern across occupations compared to the US.

Figure 3 shows that the there are 4 quadrant of occupation based on Indonesia's RCA AI Adoption compared to the US observed exposure. The first quadrant is occupations with low observed exposure and $RCA > 1$. This quadrant represent occupations that have low exposure risk in the US but are actively adopting AI usage in Indonesia. The second quadrant (II) have occupations with high observed exposure and $RCA > 1$. This quadrant represent the occupations that have high adoption and observed risk in the US and also high adoption

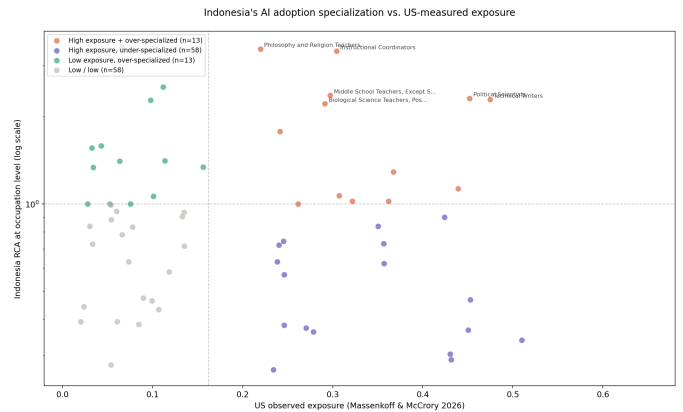


Fig. 4. Indonesia's RCA AI Adoption vs US-measured Observed Exposure

in Indonesia. This shows how AI adoption in Indonesia moves through same trajectory with adoption in the US. The third quadrant (III) is occupations with low observed exposure and $RCA < 1$. This represents the occupations that are not exposed to AI displacement risk in the US and also does not utilize AI heavily in Indonesia. The last quadrant (IV) is filled with occupations that have high observed exposure and $RCA < 1$. These are the highest AI adopting occupation in the US that does not behave on the same trajectory in Indonesia.

V. CONCLUSION

The comparative analysis on AI adoption of Indonesia reveals that 1) AI usage as shown by requests differs between countries and reveals board usage of AI in a country; 2) RCA of AI Adoption in Indonesia shows that Educational Instruction, Life, Physical & Social Science shows high RCA compared to global median while Sales, Office & Admin Support, and Computer & Mathematical still sits comparatively lower than global median of AI usage. 3) Spearman test reveals that Indonesia has different occupational AI adoption pattern with the US and occupations could be differentiated into 4 levels of quadrants of AI adoption according to RCA and US-measured observed exposure. This finding further shows the different pattern of AI adoption between Indonesia and US in occupational level. 4) Comparative analysis could provide information on the pattern of adoption of AI across countries.

This study has limitations in which 1) the AUI index is sampled data on August 2025, changes on usage pattern might occur on different time; 2) The data usage comes from Claude AI in its platform; and 3) Tasks that are below Anthropic privacy threshold are filtered out of the public release.

REFERENCES

- [1] Acemoglu, Daron. 2024. "The Simple Macroeconomics of AI." NBER Working Paper 32487, National Bureau of Economic Research, Cambridge, MA.
- [2] Acemoglu, Daron, and Pascual Restrepo, "Artificial Intelligence, Automation, and Work," *The Economics of Artificial Intelligence: An Agenda*, 2019, pp. 197–236. Edited by Ajay Agrawal, Joshua Gans and Avi Goldfarb. Chicago, USA: University of Chicago Press.
- [3] Acemoglu, Daron, David Autor, and Simon Johnson. 2023. "Can We Have Pro-Worker AI? Choosing a Path of Machines in Service of Minds." Policy Memo, MIT Shaping the Future of Work Initiative, September 19, 2023.
- [4] Appel, Ruth, Peter McCrory, Alex Tamkin, Michael Stern, Miles McCain, and Tyler Neylon. 2025. "Anthropic Economic Index Report: Uneven Geographic and Enterprise AI Adoption." September 15, 2025. <https://www.anthropic.com/research/anthropic-economic-index-september-2025-report>
- [5] Balassa, B. (1965). Trade Liberalisation and 'Revealed' Comparative Advantage. *The Manchester School*, 33(2), 99–123.
- [6] Eloundou, Tyna, Sam Manning, Pamela Mishkin, and Daniel Rock. "GPTs Are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models." arXiv:2303.10130. Preprint, arXiv, August 21, 2023. <https://doi.org/10.48550/arXiv.2303.10130>
- [7] Hollander, Myles, Douglas A. Wolfe, and Eric Chicken. *Nonparametric Statistical Methods*. Third edition. Wiley Series in Probability and Statistics. John Wiley & Sons, Inc, 2013.
- [8] Massenkoff, M., & McCrory, P. (2026, March 5). Labor market impacts of AI: A new measure and early evidence. Anthropic. <https://www.anthropic.com/research/labor-market-impacts>
- [9] U.S. Bureau of Labor Statistics. (2018). *Standard Occupational Classification Manual, 2018*. Washington, DC: Office of Management and Budget. <https://www.bls.gov/soc/2018/>